

Square Roots (Vargmool)

Aasan Hinglish Notes, Methods, Examples aur Practice Questions

Square root, square (varg) ka ulta hota hai. Is guide me concept ko simple Hinglish me examples ke saath samjhaya gaya hai, square root nikalne ke methods diye gaye hain, aur end me 22 practice questions (answer key ke saath) hain.

1. Square Root kya hota hai?

Definition Kisi number ka square root wo value hai jise khud se multiply karne par wahi number mil jaye. Ye squaring ka ulta (inverse) hai.

Symbol: Radical sign $\sqrt{\quad}$ use hota hai. Jaise $\sqrt{25}$.

Examples:

- $\sqrt{25} = 5$ kyunki $5 \times 5 = 25$
- $\sqrt{81} = 9$ kyunki $9 \times 9 = 81$
- Square: $6 \times 6 = 36$ $\rightarrow$ Square root: $\sqrt{36} = 6$

2. Perfect Squares

Definition Perfect square wo number hota hai jiska square root ek pura (whole) number ho.

Examples: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144 ...

Yaad rakho:

- Perfect squares ke square root exact whole numbers hote hain.
- 2, 3, 5, 7 jaise numbers perfect square nahi hain; inke root non-ending decimals (irrational) hote hain, jaise $\sqrt{2} = 1.414...$
- Perfect square kabhi 2, 3, 7 ya 8 par khatam nahi hota.

3. Square Root Nikalne ke Methods

(a) Prime Factorisation Number ko prime factors me todo, pairs banao, aur har pair me se ek factor lo.

- Example: $144 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 \rightarrow 2 \times 2 \times 3 = 12$, to $\sqrt{144} = 12$

(b) Long Division Bade numbers aur non-perfect squares ke liye step-by-step answer (decimals me bhi) nikalne ke liye.

(c) Estimation Do nazdeeki perfect squares dekh kar approximate value guess karo.

- Example: $\sqrt{150}$, $\sqrt{144}=12$ aur $\sqrt{169}=13$ ke beech hai, to lagbhag 12.2

4. Kaam ki Properties

- $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$
- $\sqrt{a / b} = \sqrt{a} / \sqrt{b}$
- $\sqrt{0} = 0$ aur $\sqrt{1} = 1$
- Negative number ka koi real square root nahi hota.

5. 100 se 200 ke Beech Perfect Squares

Is range me sirf paanch perfect squares hain:

$\sqrt{100} = 10$	$(10 \times 10 = 100)$
$\sqrt{121} = 11$	$(11 \times 11 = 121)$
$\sqrt{144} = 12$	$(12 \times 12 = 144)$
$\sqrt{169} = 13$	$(13 \times 13 = 169)$
$\sqrt{196} = 14$	$(14 \times 14 = 196)$

6. Practice Questions (Khud Try Karo!)

In 22 questions ko khud solve karne ki koshish karo. Answers neeche answer key me diye gaye hain.

- Q1. $\sqrt{49} = ?$
 Q2. $\sqrt{64} = ?$
 Q3. $\sqrt{81} = ?$
 Q4. $\sqrt{100} = ?$
 Q5. $\sqrt{121} = ?$
 Q6. $\sqrt{144} = ?$
 Q7. $\sqrt{169} = ?$
 Q8. $\sqrt{196} = ?$
 Q9. $\sqrt{225} = ?$
 Q10. $\sqrt{256} = ?$
 Q11. Kya 150 ek perfect square hai?
 Q12. $\sqrt{180}$ kin do whole numbers ke beech aata hai?
 Q13. $\sqrt{36} + \sqrt{64} = ?$
 Q14. Ek square park ka area 196 sq m hai. Uski side kitni hogi?
 Q15. 100 se 200 ke beech kaunse perfect squares hain?
 Q16. $\sqrt{2}$ rational hai ya irrational?
 Q17. $\sqrt{400} = ?$
 Q18. $\sqrt{1} = ?$ aur $\sqrt{0} = ?$
 Q19. Kya negative number ka real square root hota hai?
 Q20. $\sqrt{9 \times 16} = ?$ (property use karo)
 Q21. $\sqrt{625} = ?$
 Q22. $12 \times 12 = 144$, to $\sqrt{144} = ?$

7. Answer Key

- Q1: 7 ($7 \times 7 = 49$)
- Q2: 8 ($8 \times 8 = 64$)
- Q3: 9 ($9 \times 9 = 81$)
- Q4: 10 ($10 \times 10 = 100$)
- Q5: 11 ($11 \times 11 = 121$)
- Q6: 12 ($12 \times 12 = 144$)
- Q7: 13 ($13 \times 13 = 169$)
- Q8: 14 ($14 \times 14 = 196$)

- Q9: 15 ($15 \times 15 = 225$)
- Q10: 16 ($16 \times 16 = 256$)
- Q11: Nahi ($\sqrt{150} \sim 12.25$, whole number nahi)
- Q12: 13 aur 14 ke beech ($13^2 = 169$, $14^2 = 196$)
- Q13: 14 ($6 + 8 = 14$)
- Q14: 14 metre ($\sqrt{196} = 14$)
- Q15: 100, 121, 144, 169 aur 196
- Q16: Irrational
- Q17: 20 ($20 \times 20 = 400$)
- Q18: $\sqrt{1} = 1$ aur $\sqrt{0} = 0$
- Q19: Nahi (negative ka real square root nahi hota)
- Q20: 12 ($\sqrt{9} \times \sqrt{16} = 3 \times 4 = 12$)
- Q21: 25 ($25 \times 25 = 625$)
- Q22: 12